AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

COMBINED SYSTEM

CATION + ANION + DEGASSER + MIXED BED

HIGH PURITY WATER FOR INDUSTRIAL APPLICATIONS

250 LPH – 10,000 LPH

- Complete solution for demineralized water
- Compact & skid mounted design
- FRP / MSEP / MSRL options
- Efficient CO_2 removal (Degasser)
- Low maintenance & easy operation
- Suitable for boiler feed, process and utility use
- Custom engineered as per requirement



FRP COMBINED UNIT
(250 LPH – 2,000 LPH)
Skid Mounted with Frontal Piping (MPV)



MSEP COMBINED UNIT
(2,500 LPH – 5,000 LPH)
MS Vessel with Epoxy Lining, Frontal Piping



MSRL COMBINED UNIT
(7,500 LPH – 10,000 LPH)
MS Rubber Lined Vessel, Frontal Piping

APPLICATIONS


Boiler Feed Water


Pharmaceutical


Power Plant


Food & Beverage


Chemical Industry


Electronics / Semiconductor


Textile Industry


Laboratory / Research

PROCESS FLOW DIAGRAM



TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Typical Inlet (Raw Water)	Typical Outlet (DM Water)
pH	–	6.5 – 8.5	6.5 – 7.0
Total Hardness (as $CaCO_3$)	mg/L	50 – 500	< 0.1
Calcium Hardness	mg/L	20 – 300	< 0.1
Magnesium Hardness	mg/L	10 – 200	< 0.1
Alkalinity (as $CaCO_3$)	mg/L	50 – 400	< 5
Dissolved CO_2	mg/L	10 – 50	< 1
TDS	mg/L	100 – 2,000	< 10
Silica (as SiO_2)	mg/L	5 – 50	< 0.02
Conductivity	µS/cm	200 – 3,000	< 1

KEY FEATURES & BENEFITS

- Integrated Cation, Anion, Degasser and Mixed Bed
- High purity DM water with low conductivity
- Efficient CO_2 removal by air stripping degasser
- Available in FRP, MSEP (Epoxy Lined) and MSRL (Rubber Lined) vessels
- Compact skid mounted with frontal piping
- Automatic / Manual operation options
- Suitable for flow range 250 LPH to 10,000 LPH
- Low operating cost and easy maintenance
- Pre-engineered and factory tested
- Customizable design as per raw water and quality requirement

INTERNALS & MEDIA


Cation Resin (SAC)


Anion Resin (SBA)


Mixed Bed Resin (Pre-mixed)


FRP Vessel


MSEP Vessel (Epoxy Lining)


MSRL Vessel (Rubber Lining)

TECHNICAL SPECIFICATIONS & DIMENSIONS (COMBINED UNIT – CATION + ANION + DEGASSER + MIXED BED)

Model	Flow Rate (LPH)	Vessel Type (FRP / MSEP / MSRL)	Cation Bed			Anion Bed			Degasser Tower		Mixed Bed			Skid Size (Approx.) (L x W x H) (mm)	Inlet / Outlet Size (NB)
			Dia (mm)	HOS (mm)	Resin Qty (Lit)	Dia (mm)	HOS (mm)	Resin Qty (Lit)	Dia (mm)	Height (mm)	Dia (mm)	HOS (mm)	Resin Qty (Lit)		
AQU-CADM-0.25	250	FRP	200	1,200	25	200	1,200	25	250	2,000	200	1,200	25	2,200 x 1,000 x 2,200	15
AQU-CADM-0.5	500	FRP	250	1,500	50	250	1,500	50	300	2,500	250	1,500	50	2,800 x 1,200 x 2,600	20
AQU-CADM-1.0	1,000	FRP / MSEP	350	1,800	100	350	1,800	100	400	3,000	350	1,800	100	3,800 x 1,500 x 3,100	25
AQU-CADM-2.0	2,000	MSEP	450	2,500	200	450	2,500	200	500	3,500	450	2,500	200	4,500 x 1,800 x 3,800	40
AQU-CADM-3.0	3,000	MSEP	500	3,000	300	500	3,000	300	600	4,000	500	3,000	300	5,200 x 2,000 x 4,300	50
AQU-CADM-5.0	5,000	MSEP	650	3,500	500	650	3,500	500	700	5,000	650	3,500	500	6,500 x 2,200 x 5,300	65
AQU-CADM-7.5	7,500	MSRL	800	4,000	750	800	4,000	750	900	6,000	800	4,000	750	7,800 x 2,500 x 6,300	80
AQU-CADM-10.0	10,000	MSRL	1,000	4,500	1,000	1,000	4,500	1,000	1,200	7,000	1,000	4,500	1,000	9,000 x 3,000 x 7,500	100

Note: Dimensions are indicative and can be customized as per site conditions and water quality. HOS – Height of Straight Shell.

OPTIONAL FEATURES

- Online pH, Conductivity and TDS monitoring
- Automatic regeneration system (for individual units)
- SS / UPVC / CPVC piping options
- Skid with chemical dosing (for pH correction)
- DM water storage tank and transfer pump
- Custom design as per client requirement

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

Office & Factory :
Sr. No. 165/2/9, Alankapuram Road, Opposite Yashobhumi Society, Adarsha Nagar, Charholi Budruk, Pimpri-Chinchwad, Maharashtra – 412105, India.

+91 9096645119 / +91 9622554445
ceo@aquenfluid.com
www.aquenfluid.com

CLEANER WATER GREENER TOMORROW

1. FRP COMBINED UNIT – 1,000 LPH

CATION + ANION + DEGASSER + MIXED BED (FRP VESSELS)



2. MSEP COMBINED UNIT – 5,000 LPH

CATION + ANION + DEGASSER + MIXED BED (MSEP VESSELS WITH EPOXY LINING)



3. MSRL COMBINED UNIT – 10,000 LPH

CATION + ANION + DEGASSER + MIXED BED (MSRL VESSELS WITH RUBBER LINING)



4. COMPACT INTEGRATED UNIT – 2,500 LPH

CATION + ANION + DEGASSER + MIXED BED (FRP / MSEP HYBRID – SKID MOUNTED)

